

I- 1

- xml approach
- class
- xml
- ClassPathXmlApplicationContext

xml configuration approach

I- 2

SteroType annotation

```
@component ,@service,@Controller
@RestController,@Repository
```

```
@Required ,@Autowired,
@Qualifier , @Scope , @lazy
@DependenOn , @value,
@propertySource
```

I- AnnotationConfigApplicationContext

I- 3

Now think about we have a class

```
class A {
    File file;
    setter for injection
}
```

How to tell the IOC container about this class so that Ioc container can able to perform all the operations

application-context.xml

```
<bean id="a" class="package.A">
  <property name="file" ref="file" />
</bean>
```

Text

```
File file=new File("E://download/read.txt");
```

ApplicationContext context= new ClassPathXmlApplicationContext("application-context.xml");

I- 4

Can we do the same thing without xml

Annotation

```
@component
class A{
    @Autowired File file;
}
```

Text

```
class A{
    B b;
    class B {}
}
id="a" class="A"
id="b" class="B"
```

Text

```
class A{
    @Component
    class B{
    }
}
```

5. Combination of Annotation and xml

```
@component
class A
{
    @Autowired
    File file;
    setter
}
```

application-context.xml

```
<bean id="file" class="IO.File">
  <constructor-args location="" />
</bean>
<context:component-scan base package="" />
```

ApplicationContext context=new ClassPathXmlApplicationContext(context(" "));

IF THERE IS THE SOURCE CODE PRESENT

- ANNOTATION

IF THERE IS NO SOURCE CODE

- Xml Approach

Spring had provided one alternate thing which is

JAVA CONFIGURATION APPROACH

ApplicationContext context=new Annotation_____

6

```
@component
Class Toy {
    @value("Iron Man")
    private String toyName;
    @value("10000")
    private int priceOfToy;
    setter & toString();
}
```

xml

```
bean
id="toy" class="package.TOY"
<property name="toyname"
value="IronMan"/>
<property name="priceOfToy"
value="10000"/>
bean close</bean>
```

I need one more set of bean with different value

can we create one more object with different value

```
<bean id="toy1" class="package.Toy">
  <property name="toyName"
value="CaptainAmerica"/>
  <property name="priceOfToy"
value="10000"/>
</bean>
```

spring bean definition

ApplicationContext context= new ClassPathXmlApplicationContext();

context.getBean("toy1",Toy.class); → spring bean

7. I don't want to work with xml approach then can we shift to the annotation

8. approach in the same example

If we want to create multiple set of values then with annotation approach then the issue is there is only one class present

I- Problem with the annotation approach is

1. If we don't have the source code and our class is dependent on some other class whose source code is not present then only the annotation will not work
2. We cannot create single object with the multiple value with the help of annotation only
3. Suppose if tomorrow id we are going to use some other framework instead of spring again all the source code need to be change as all the annotation which spring supports are present there in the source code directly